



Improved Nu number correlations for gas flow in monolith reactors using temperature-dependent fluid properties

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ARTICLE INFO

Keywords:

CFD
Multiscale
Monolith
Nusselt
Variable properties

ABSTRACT

This paper reports an investigation of heat and mass transfer coupled with fluid flow inside monolith honeycomb substrates. Many important aspects of multiscale models of a monolith are investigated. Three common geometrical representations of monolith channels are tested, and their advantages and disadvantages are analysed. A detailed computational model and a large set of computational experiments, covering a broad range of substrate void fractions, channel Reynolds numbers and heating rates is used. Results assuming constant and temperature-dependent fluid properties are reported and compared. At a high heating rate, assuming constant fluid properties leads to a significant error in the estimation of Nusselt and Sherwood numbers in the entry region. At a constant wall temperature and a high heating rate, there is a minimum in the Nusselt curve that classical models are not able to describe. Hence, a novel correlation that shows excellent agreement with channel data is presented together with a methodology to fit it. Additionally, empirical models to estimate the placement and value of the minimum Nusselt are proposed. These results contribute significantly to the improvement of lumped parameter multiscale models of monolith catalytic reactors.

1. Introduction

Monolith honeycombs are extensively used as catalytic reactor substrates in industrial applications, and especially in automotive catalytic converters [1–6], which is the application analysed in this paper. Full scale models of such reactors are necessary to investigate important variables such as the flow distribution, heating time, pressure drop and conversion efficiency. A single monolith can have thousands of channels running in parallel and modelling each one of them individually is computationally prohibitive. As an alternative, current reactor scale models use an an-isotropic continuum porous medium to approximate the substrate. When using a continuum porous medium model, channel scale spatial discretisation is lost, and the coupling of the fluid and solid phases is achieved using heat and mass transfer coefficients. In effect, the channel level is treated as a lumped parameter model. It is evident that the correct selection of the heat and mass transfer coefficients can be critical in capturing the reactor performance accurately, especially in the entrance region. Appropriate values of Nusselt (Nu) and Sherwood (Sh) numbers can be determined from detailed 2D or 3D simulations of a single channel, provided that the correct operating conditions are used.

Nu correlations are used not only for heat transfer, but also for mass transfer. Using the law of gases and kinetic theory it can be shown that

the Lewis number (Le), which is the ratio of the convective heat transfer and convective mass transfer, remains approximately constant over a wide range of temperatures for gases. Hence, the same correlations for Nu apply for Sh. Graetz [7] made the first significant contribution to convective heat transfer in circular ducts. After that, there have been many investigations of heat and mass transfer inside of monolith channels, most of them based on the work of Kays and London [8] for flow in pipes. Several models have been proposed, each of them subject to their own limitations and assumptions. A useful correlation for Nu and Sh must, as a minimum, describe consistently the entrance region, converge asymptotically to the analytic value in the fully developed zone and be applicable over a wide range of conditions. A mathematical expression adopted by many authors has the following form [9–11]:

$$Nu = Nu_{\infty} [1 + B(Gz)^n \exp(-C/Gz)] \quad (1)$$

$$Sh = Sh_{\infty} [1 + B(Gz_m)^n \exp(-C/Gz_m)] \quad (2)$$

where

$$Gz = \frac{D}{x} RePr \quad (3)$$

$$Gz_m = \frac{D}{x} ReSc \quad (4)$$

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Eqs. (1) and (2) can be used either for a constant wall flux or a constant wall temperature (or concentration for Sh) using the appropriate B , C and n , and asymptotic values (Nu_∞ and Sh_∞). Model parameters reported by several authors are summarised in [12]. When temperature gradients are small, constant fluid properties can be assumed without a significant loss of accuracy. However, in many engineering situations, from nuclear reactors [13] to automotive catalytic converters [14], changes of temperature in both axial and radial direction are sufficiently large to affect the flow field, especially due to changes in μ and ρ [15]. In such cases, the fluid properties must be considered temperature-dependent, a fully coupled set of transport equations must be solved, mathematical superposition is not allowed and a general solution in the form of Eq. (1) is not possible. In a monolith the problem is even more complex because of the influence of the reduction of the flow area entering the substrate on the inlet velocity profile of the channels. Shumway and McEligot [16] compared the entrance region Nu for laminar flow of a Newtonian fluid in annuli considering temperature-dependent fluid properties, finding substantial differences when comparing the results to those assuming constant properties. Nobrega et al. [17] reported similar results for a viscoelastic fluid in a circular duct. Hayes et al. [12] also showed that the use of temperature-dependent physical properties gives different results from the constant property case, and also demonstrated the effect of adding a reservoir prior to the channel. Unfortunately, the previously mentioned results are not directly applicable to a monolith or cover a limited set of conditions only.

There are many publications concerning convective heat transfer in monoliths, each based on different sets of assumptions, especially regarding the channel geometry and fluid properties. This variety imposes difficulties when selecting the correct model for every occasion. The motivation of this paper is to provide readers with a deeper understanding of the influence of every assumption on the results when modelling a monolith channel and thus to allow them to make a simpler and better model selection. The main goal of this paper is to contribute substantial new evidence of the influence of using constant or temperature-dependent fluid properties and geometrical simplifications on the convective transport of gas flow through monolith channels. Those are open questions among the community that, to the best of our knowledge, have not been addressed in sufficient detail in a systematic way. A wide range of conditions is investigated and new correlations are proposed.

2. Computational model

2.1. Channel models

In this paper we investigate a monolith with circular cross-section channels, which is a commonly assumed geometry for washcoated channels [18]. When modelling monolith channels, three geometrical approaches can be identified, as illustrated in Fig. 1:

- Channel alone (or straight) in Fig. 1a: The simplest approach. The domain runs from the inlet to the outlet of the channel. A flat or fully developed inlet velocity profile is usually assumed.
- Channel and a reservoir (or reservoir) in Fig. 1b: A section with the same diameter as the channel is added prior to it. The outer boundaries of the reservoir are treated as symmetry planes. The purpose of the reservoir is to account for the radial and axial diffusion at the inlet of the channel, looking for a more realistic velocity inlet profile.
- Channel with a contraction inlet (or contraction) in Fig. 1c: An open space is added before the channel. The height of the open space depends on the substrate void fraction. This approach accounts not only for the radial diffusion at the inlet of the channels, but also for a physically consistent velocity of flow entering a monolith.

For simplicity, in the rest of the paper, the three described geometries are called “straight”, “reservoir” and “contraction” respectively. Among the three approaches, the channel with the contraction is the most realistic one. It requires more computational effort, however, there is substantial evidence about the effect of such a contraction locally on the inlet velocity profile and globally on the total pressure drop [19,20]. Although a channel with a contraction is used in this paper, the two other approaches are also investigated to quantify and to illustrate the magnitude of the error in the results when using these simplified shapes. In the three cases, the diameter of the channel (D) was 1 mm and its length (L) was 75 mm. The extension of the reservoir (L_1) was 10 mm and its diameter was the same as that of the channel. For the channel with the contraction, the inlet diameter (D_1) was manipulated to obtain the desired reduction rate for every case and the length of the open section was also 10 mm. The domain was treated as axial symmetrical.

2.2. Flow model

In many monolith applications, the channel Re is sub-critical and the flow is laminar. There is evidence that in some cases, for example, in automotive catalytic converters, upstream turbulence enters the substrate and then decays along it [21–23]. To describe such a flow in transition accurately requires large computational effort; hence, the incremental effect of the upstream turbulence will be addressed in a future investigation [24]. Therefore, in this paper the flow regime was assumed to be laminar. The steady transport equations of mass and momentum are [25]:

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (5)$$

$$\frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \frac{\partial u_i}{\partial x_i} \delta_{ij} \right] \quad (6)$$

The inlet of the domain was treated as a velocity inlet with a flat velocity profile and a constant fluid temperature. The walls of the channel were set as no-slip walls, with a fixed temperature or fixed wall heat flux, depending on the case analysed. The outlet of the domain was set as a pressure outlet.

2.3. Fluid properties

We now compare the use of constant and temperature-dependent transport properties for the convective heat transfer and the selection of a reference temperature in lumped models. Investigations of the results of varying the properties one at a time have been partially addressed in the literature, establishing that the most important properties that modify Nu are the density and the thermal conductivity [16,17]. That academic exercise provides useful information, but it is non-physical. In this paper the fluid was considered to be air; therefore, the results are applicable to gases with a Pr close to 0.7. Two scenarios described below were considered. In both the density was temperature-dependent and was calculated using the ideal gas law, meanwhile, for the other properties:

- Constant properties: μ , k and C_p are constant and are evaluated at the inlet temperature.
- Variable properties: μ , k and C_p are all calculated at the local fluid temperature.

The relationship between fluid properties and temperature is often represented by a polynomial. However, to minimise approximation errors, we used fundamental expressions when possible, such as the kinetic theory and the ideal gas law. Both are available in many commercial codes and can be handled with the current computational power. In the two scenarios, the density was calculated with the

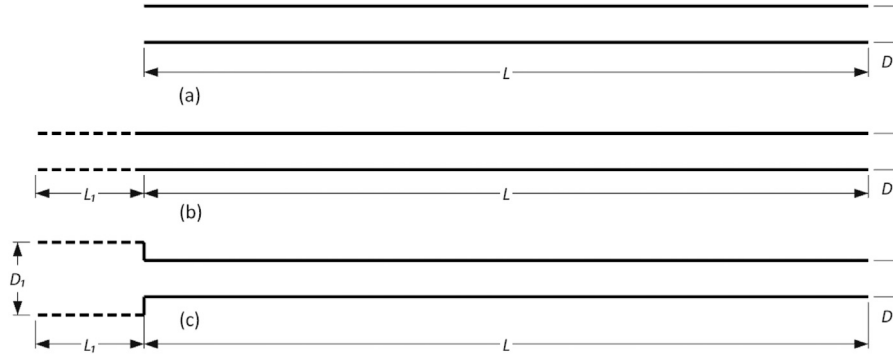


Fig. 1. (a) channel alone, (b) channel with a reservoir and (c) channel with the inlet contraction. $L = 75$ mm, $D = 1$ mm, $L_1 = 10$ mm. D_1 was case-dependent.

Table 1
Parameters for C_p in J/kg K.

Parameter	$100 < T, K < 1000$	$1000 < T, K < 3000$
a_0	1.161×10^3	-7.069×10^3
a_1	-2.3688	3.371×10^1
a_2	1.486×10^{-3}	-5.813×10^{-2}
a_3	-5.035×10^{-5}	5.422×10^{-5}
a_4	9.929×10^{-8}	-2.937×10^{-8}
a_5	-1.111×10^{-10}	9.238×10^{-12}
a_6	6.540×10^{-14}	-1.566×10^{-15}
a_7	-1.574×10^{-17}	1.112×10^{-19}

ideal gas law, to account for local changes in pressure. When using temperature-dependent properties, C_p was estimated by an 8th order piece-wise polynomial, meanwhile, the kinetic theory of gases was used for μ and k . The exact mathematical expressions were:

$$\rho \text{ (kg/m}^3\text{)} = \frac{p}{R_g T / M_w} \quad (7)$$

$$C_p \text{ (J/kg K)} = a_0 + a_1 T + a_2 T^2 + a_3 T^3 + a_4 T^4 + a_5 T^5 + a_6 T^6 + a_7 T^7 \quad (8)$$

where the parameters are [25] (see Table 1).

$$k \text{ (W/m K)} = \frac{15}{4} \frac{R_g}{M_w} \mu \left[\frac{4}{15} \frac{C_p M_w}{R_g} + \frac{1}{3} \right] \quad (9)$$

$$\mu \text{ (Pa s)} = 2.67 \cdot 10^{-6} \frac{\sqrt{M_w T}}{\sigma^2 \Omega_\mu} \quad (10)$$

For μ , the characteristic length (σ) was 3.711 and Ω_μ is a function of the reduced temperature that can be found in [25,26]. Equations (7) to (10) are utilised as they are implemented in the commercial solver Fluent v18.2 [27], which is broadly used in the literature.

2.4. Calculation methodology

Convective heat transfer throughout a monolith channel is analysed either at a constant wall temperature or at a constant wall heat flux. Nu is defined as:

$$\text{Nu} = \frac{hD}{k} \quad (11)$$

k was evaluated at the local wall temperature, meanwhile, the other properties of the fluid, namely, viscosity and heat capacity, were evaluated at the local mixing-cup one, which were the same as those at the inlet when assuming constant properties. The convective coefficient was obtained from the temperature field according to:

$$h(T_w - T_b) = -k \frac{dT}{dr} \Big|_{y=R} \quad (12)$$

The temperature gradient was evaluated at the wall ($y = R$). Especial attention should be paid to the computation of the temperature gradient. At least a second order of accuracy approximation should be used. The mixing-cup, or bulk, temperature (T_b) was defined in the usual way:

$$T_b = \frac{\int_0^R u_x \rho C_p T r dr}{\int_0^R u_x \rho C_p r dr} \quad (13)$$

Nu was plotted against a non-dimensional axial position represented by the inverse of the Graetz number (Gz). To compare different options of reference temperature commonly found in the literature, Gz was calculated in two ways. The first one used the fluid properties evaluated at the local temperature and the second evaluated at the inlet one. The two mathematical expressions differ in the subscript i , which stands for inlet, and are the following:

$$1/\text{Gz} = \frac{x^+}{\text{Pr}} = \frac{x}{D} \frac{1}{\text{RePr}} \quad (14)$$

$$1/\text{Gz}_i = \frac{x_i^+}{\text{Pr}_i} = \frac{x}{D} \frac{1}{\text{Re}_i \text{Pr}_i} \quad (15)$$

When using temperature-dependent properties, the mixing-cup temperature was used to obtain a representative average of all of the radially distributed values of every channel cross-section. Another important variable often analysed in heat transfer problems is the loading temperature or heating rate, which for a channel with a constant wall temperature is defined as:

$$\tau = \frac{T_{\max}}{T_{\min}} \quad (16)$$

Table 2 shows the list of all of the computational experiments involved in this paper. This investigation covered channel Re from 50 to 600 and heating rates (τ) from 1.1 to 3.0. Those ranges are sufficient to study the effect of the two variables in the Nu vs. Gz curves and cover a wide range of applications.

3. Results

3.1. Grid independence and validation

The quality of the computational grid was investigated by comparing the numerical results to solutions available in the literature. We started by modelling a channel without a reservoir and with a flat inlet velocity profile. Once having a validated solution, the same element size and refinement were applied to the reservoir and the open space before the channel in the other two geometries; hence, the meshes for the three geometries are homogeneous and have an identical resolution inside the channel. A section of the mesh is shown in Fig. 2.

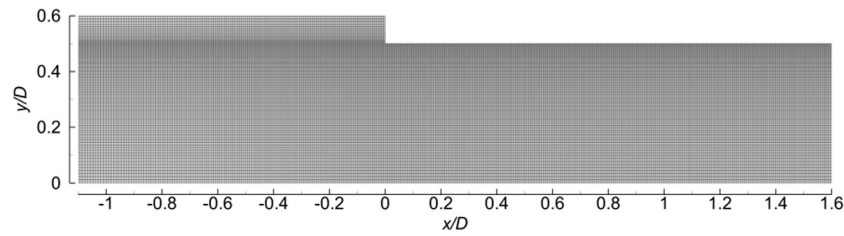


Fig. 2. Section of the computational grid for a channel with a reduction rate of 0.67. The mesh has 80 radial elements inside the channel.

Table 2

List of computational experiments.

Run	Re	T_i (K)	T_w (K)	H (W/m ²)	Geometry ^a	ϕ	Flow prop. ^b
1	150	300	330	–	Straight	–	Variable
2	150	300	600	–	Straight	–	Variable
3	300	300	330	–	Straight	–	Variable
4	300	300	600	–	Straight	–	Variable
5	50	300	330	–	Straight	–	Variable
6	50	300	330	–	Straight	–	Constant
7	600	300	330	–	Straight	–	Variable
8	600	300	330	–	Straight	–	Constant
9	50	300	600	–	Straight	–	Variable
10	50	300	600	–	Straight	–	Constant
11	300	300	600	–	Straight	–	Variable
12	300	300	600	–	Straight	–	Variable
13	300	300	600	–	Reservoir	–	Variable
14	300	300	600	–	Contraction	0.67	Variable
15	300	300	–	500	Straight	–	Variable
16	300	300	–	500	Reservoir	–	Variable
17	300	300	–	500	Contraction	0.67	Variable
18	300	300	600	–	Contraction	0.45	Variable
19	300	300	600	–	Contraction	0.80	Variable
20	300	300	–	500	Contraction	0.45	Variable
21	300	300	–	500	Contraction	0.80	Variable
22	50	300	600	–	Contraction	0.67	Variable
23	150	300	600	–	Contraction	0.67	Variable
24	450	300	600	–	Contraction	0.67	Variable
25	600	300	600	–	Contraction	0.67	Variable
26	150	300	330	–	Contraction	0.67	Variable
27	150	300	400	–	Contraction	0.67	Variable
28	150	300	500	–	Contraction	0.67	Variable
29	150	300	600	–	Contraction	0.67	Variable
30	50	300	–	1000	Contraction	0.67	Variable
31	150	300	–	1000	Contraction	0.67	Variable
32	150	300	–	1000	Contraction	0.67	Variable
33	450	300	–	1000	Contraction	0.67	Variable
34	600	300	–	250	Contraction	0.67	Variable
35	150	300	–	2500	Contraction	0.67	Variable
36	150	300	–	4000	Contraction	0.67	Variable
37	150	300	900	–	Contraction	0.67	Variable
38	150	300	750	–	Contraction	0.67	Variable

^aAccording to Section 2.1.

^bAccording to Section 2.3.

Taking advantage of the cross-section of the channel, axial-symmetry was used. For the channel, two fully orthogonal meshes of 492 164 and 1 968 656 quadrilateral control volumes were tested. The meshes contained 80 and 160 radial elements respectively. The monitored variables were the Nu vs. 1/Gz curve, and particularly Nu in the fully developed region. Results for a constant wall temperature case are shown in Fig. 3.

The two meshes gave the same Nu vs. 1/Gz curves in the entrance region and agreed with literature values in the entrance and fully developed zones. Equivalent results were observed for a constant wall heat flux; hence, the mesh with 492 164 control volumes inside the channel was used in the rest of the study. A low heating rate was used in the validation case, to avoid large changes introduced by the variable fluid properties, which is discussed later in this paper.

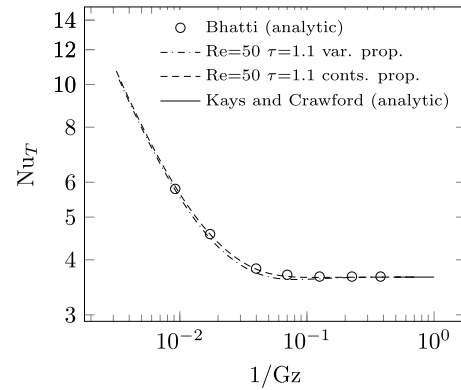


Fig. 3. Grid investigation. The fully developed Nu for circular pipes from Kays and Crawford [28] is shown as a reference. Bhatti solution, as cited in [28], assumed constant fluid properties [29].

3.2. Selection of a reference temperature

The Nusselt number in the entrance region is usually plotted against a representation of the dimensionless axial distance from the inlet. The most common representation is the inverse Graetz number, which is robust because it accounts for changes in both Re and Pr [30]. When calculating the value of 1/Gz for the case of temperature-dependent physical properties, the temperature used when computing Nu, Re and Pr affects the results. Common choices are to use the inlet temperature, a film temperature or the local fluid temperature. The film temperature is an average that can be defined in different ways, but is usually the average of the wall and the mean bulk temperature. Fig. 4a shows the change in Re and Pr as a function of distance from the inlet for a straight channel with a constant wall temperature higher than the inlet, with both Re and Pr calculated at the local mixing-cup temperature. Changes in both variables along with the axial coordinate can be noticed, although, the variations in Pr very low in magnitude, changes in Re vary with the heating rate.

The product of Re and Pr, the Peclet number for heat transfer (Pe_H), changes with the temperature according to the results shown in Fig. 4b. On this graph the ratio of RePr computed at local temperature to the values computed at inlet temperature is shown as a function of a dimensionless axial distance. Clearly, the error introduced by using the inlet temperature is case specific, depending on both the Re and heating rate, with possible differences as high as 50% or more. The effect of using the inlet temperature to develop the Nu vs. 1/Gz curve is to shift the apparent curve to the right, leading to a different h value. That would also affect the estimation of the thermal and hydraulic entry length. An analysis using a film temperature would also give a similar result.

3.3. Constant vs. temperature-dependent fluid properties

This section briefly revisits the error in the entrance length Nu when assuming constant fluid properties. This point has been addressed in

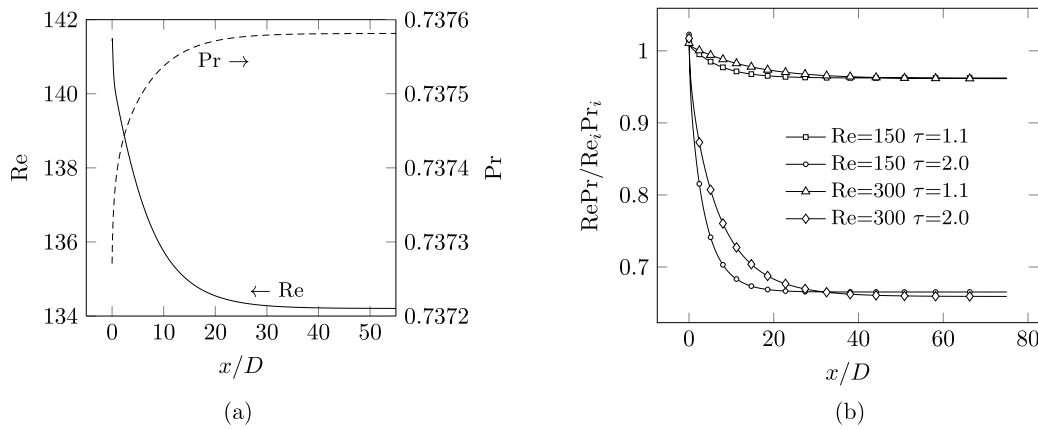


Fig. 4. Example of the change in Re and Pr through a straight channel with a flat inlet velocity profile at $T_i = 300$ K and a constant wall temperature. All the curves were obtained using temperature-dependent fluid properties and correspond to runs 1 to 4 of Table 2.

the literature. However, experiments are commonly carried out over a limited set of conditions and the literature is particularly sparse in terms of the heating rate range. This part of the investigation follows the fashion of a fully orthogonal 2^2 factorial design to test the influence of the channel Re and heating rate on the displacement of the Nu vs. $1/Gz$ curve when assuming constant and temperature-dependent fluid properties. Values of 50 and 600 were considered low and high levels for the channel Re and 1.1 and 2.0 for the heating rate. A straight channel with a flat inlet profile and a constant wall temperature was used. Results are shown in Fig. 5.

Fig. 5 is organised in a way that shows Nu vs. $1/Gz$ curves for the following combinations: a low Re and a low τ , a low Re and a high τ , a high Re and a low τ , and a high Re with a high τ . Comparing Fig. 5 a and c shows that for a low heating rate the entrance Nu is not significantly affected by the assumption of constant properties over a wide range of Re. This effect occurs because at a low heating rate the temperature change in the inlet region is minor, thus the use of inlet property values does not lead to significant error. It is also intuitive that for an increase of Re, all the other variables being equal, the temperature change in the inlet region is reduced. On the other hand, at a high heating rate, the gap between the curves using constant and variable properties is no longer negligible. Assuming constant fluid properties always overestimates Nu, especially at a high Re. It should be mentioned that the minimum in Fig. 5b and d has been reported previously in the literature [16,17,31]. It is always present, but commonly, its value is comparable to the asymptotic one and it is not appreciable in the curve. That phenomenon, which affects visibly the entrance length, is not captured correctly when constant fluid properties are assumed. As shown in Fig. 6c, the minimum in the Nu curve is present regardless of having or not a reduction of the flow area entering the substrate. In this paper, similarly to the results in More [31], the minimum in the Nu curve appears as soon as the fluid properties are set to be temperature-dependent. According to More, when setting the fluid properties to be variable one at the time, the one that show the largest influence on the formation of the minimum is the fluid viscosity.

3.4. Effect of shape simplifications

This section addresses the differences in Nu vs. $1/Gz$ curves observed for various inlet geometries. The shapes tested were those described in Section 2.1. Variable fluid properties and an inlet temperature of 300 K were used in all the computational experiments of this section. Results for both Nu_T and Nu_H are shown in Fig. 6.

According to Fig. 6a, adding a reservoir effectively improves the radial diffusion at the inlet of the channel, because the velocity profile is resolved rather than imposed. However, that is not sufficient to

capture the change of the direction of the flow due to the reduction of the open area entering the substrate (see Fig. 6b). That has been reported for iso-thermal and oscillating flow [19,20], and now its effect on the convective heat transfer is shown. It can be seen in Fig. 6c and d, that both the channel alone and when adding a straight reservoir overestimate Nu over a significant part of the entrance length.

3.5. Effect of void fraction

Since the reduction of the flow area at the entrance of the substrate affects the development of the boundary layer, the Nu curves produced for different substrate void fractions should be analysed. Three void fractions were selected to cover most cases, including highly loaded dual layer washcoat [32]. The results are shown in Fig. 7.

According to Fig. 7, for both a constant wall temperature and a constant wall heat flux, changing the void fraction modifies Nu at the entry region. However, the magnitude of the changes is much lower than the one resulting from neglecting the reduction of the flow area. It seems that as long as the contraction exists, its influence depends only slightly on the reduction rate [20,33], over any realistic range.

3.6. Parametric investigation of the effect of Re, τ and H

Having demonstrated the importance of using variable physical properties and the correct inlet geometry, and showing the conceptual relationship between Re and τ or H , we now show the results of a parametric study. In all of the cases the channel with a contraction at the inlet was used, along with variable physical properties. Because of the bi-directional coupling between the heat transfer and the flow field, the resulting Nu vs. $1/Gz$ curves are case dependent, rather than being a unique one. The results of the study are shown in Fig. 8.

It is clear from Fig. 8 that the effect of changing Re, τ and H is to alter the shape of the curve in the developing region. The first observation is that even for a relatively high heating rate (large τ , Fig. 8a or large H , Fig. 8c) the effect of changing Re over a wide range has a relatively minor effect that is concentrated very close to the inlet. This effect is primarily due to the contraction of the flow at the inlet. In cases where the light-off occurs very close to, or at, the inlet this difference might play a role. On the other hand, for a constant wall temperature and a fixed value of Re, the effect of the heating rate is quite significant (Fig. 8b). Although, the slope of the curve is approximately the same for all values of τ , the curve shifts to the left as τ increases. The value of the minimum observed also becomes significantly lower as τ increases, reaching a value of 16% below the final asymptotic one (Run 29). Interestingly, for the cases of a constant wall heat flux with a fixed Re, an increase in H has a much more minor effect on the curve. The minimum on the Nu vs.

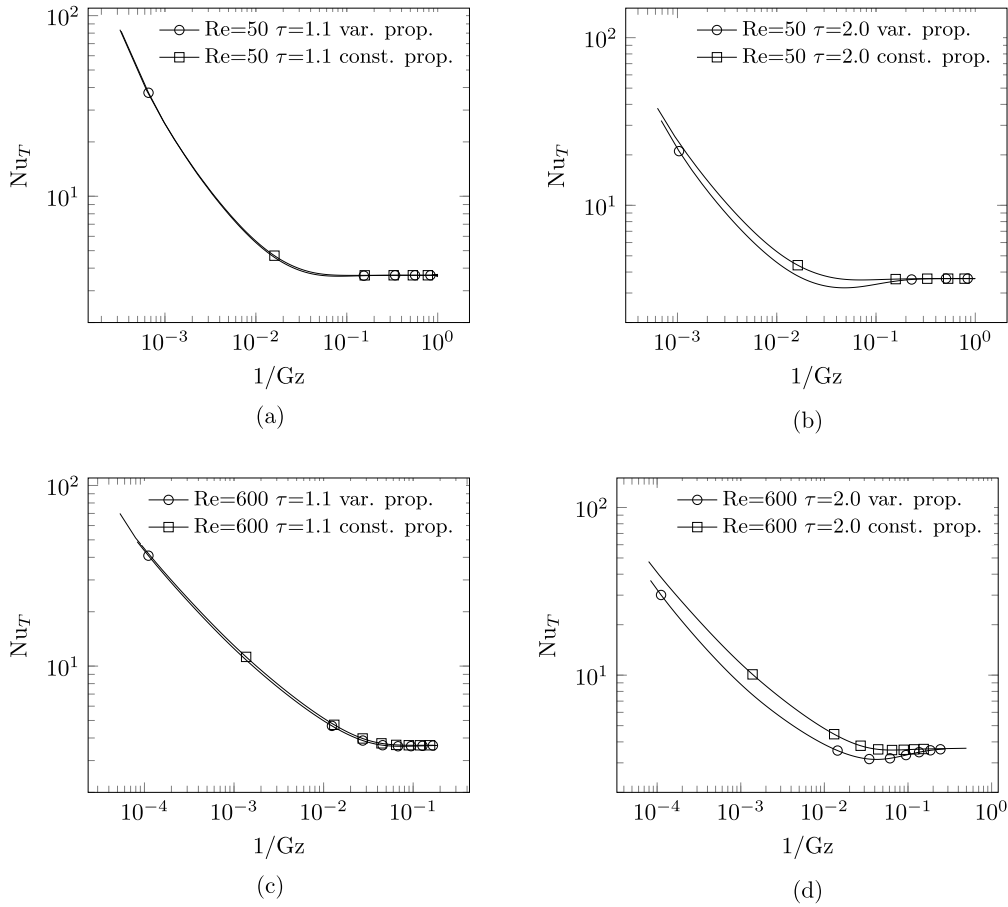


Fig. 5. Comparison of Nu vs. $1/Gz$ curves obtained using constant and variable fluid properties at different combinations of Re and τ . Runs 5 to 12 of Table 2.

$1/Gz$ plot remains, especially at large value of H , however, it is also less pronounced than for the case of constant wall temperature. The overall conclusion from this short parametric study is that the use of a single Nu vs. $1/Gz$ curve for all the cases might lead to appreciable error under some circumstances.

3.7. Proposed correlations for Nu and Sh vs. $1/Gz$ in monolith channels

In this section we propose some corrections for the Nu in the entry region for the case with the inlet contraction and variable physical properties.

We consider first the case of a constant wall heat flux. For a constant wall heat flux (Nu_H), a single curve fitted to classical models can be used without much sacrifice of accuracy. Runs 30 to 36, covering Re from 50 to 600 and H from 0.25 kW/m² to 4 kW/m², were fitted simultaneously to Eq. (1). The overall R^2 was 0.9901. The data and fitting line can be seen in Fig. 9a.

For the situation with a constant wall temperature, Nu_T , at a high heating rate classical models do not apply, because of the presence of the minimum in the curve. A simple solution often found in heat transfer is to use a piece-wise function. The curves with a minimum can be decomposed into a strictly decreasing section and an S-shaped section. The first part can be fit to a classic model with the mathematical structure of Eq. (1), using B , C and n from Table 4. For the second part, we selected a sigmoid function extensively used in machine learning. Fig. 10a shows an example of the two functions fitted to a dataset.

The two parts of the curve can be reasonably described by blending the two functions. Taking advantage of the behaviour of the curves before and after they intercept, a simple way to combine them is to

Table 3

List of parameters for Nu_T curves at several τ .

Run	τ	$Nu_{T_{min}}$	Gz_*^{-1}	B	C	n	a	b
27	1.33	3.4546	0.05442	0.05613	67.47	0.4860	14.43	1.3360
28	1.66	3.2710	0.04542	0.04489	75.35	0.5053	15.77	0.1363
29	2.00	3.1365	0.03973	0.03750	82.41	0.5207	15.54	0.1205
38	2.50	3.0167	0.03591	0.03049	92.98	0.5376	15.80	0.1100
37	3.00	2.9249	0.03286	0.02610	102.8	0.5497	15.19	0.1353

pick the highest between the two predicted values, which leads to the following expression:

$$Nu_T = \max(Nu_T^{dec}, Nu_T^{sig}) \quad (17)$$

where

$$Nu_T^{dec} = Nu_{T_{min}} \left[1 + B(Gz)^n \exp\left(-\frac{C}{Gz}\right) \right] \quad (18)$$

$$Nu_T^{sig} = \frac{Nu_{T_{\infty}}}{Nu_{T_{min}}} \left[\left(\frac{1}{1 + e^{-\zeta}} \right) + Nu_{T_{min}} - 1 \right] \quad (19)$$

$$-\zeta = -a(Gz_*^{-1} - Gz_*^{-1}) + b \quad (20)$$

The empirical parameters involved are B , C and n for Eq. (18), and a and b for Eq. (19). Gz_*^{-1} in Eq. (20) is the axial position of the minimum of the curve. Values for them together with the minimum Nu observed for several heating rates are reported in Table 3. Excellent agreement was achieved when the same methodology was applied to different heating rates.

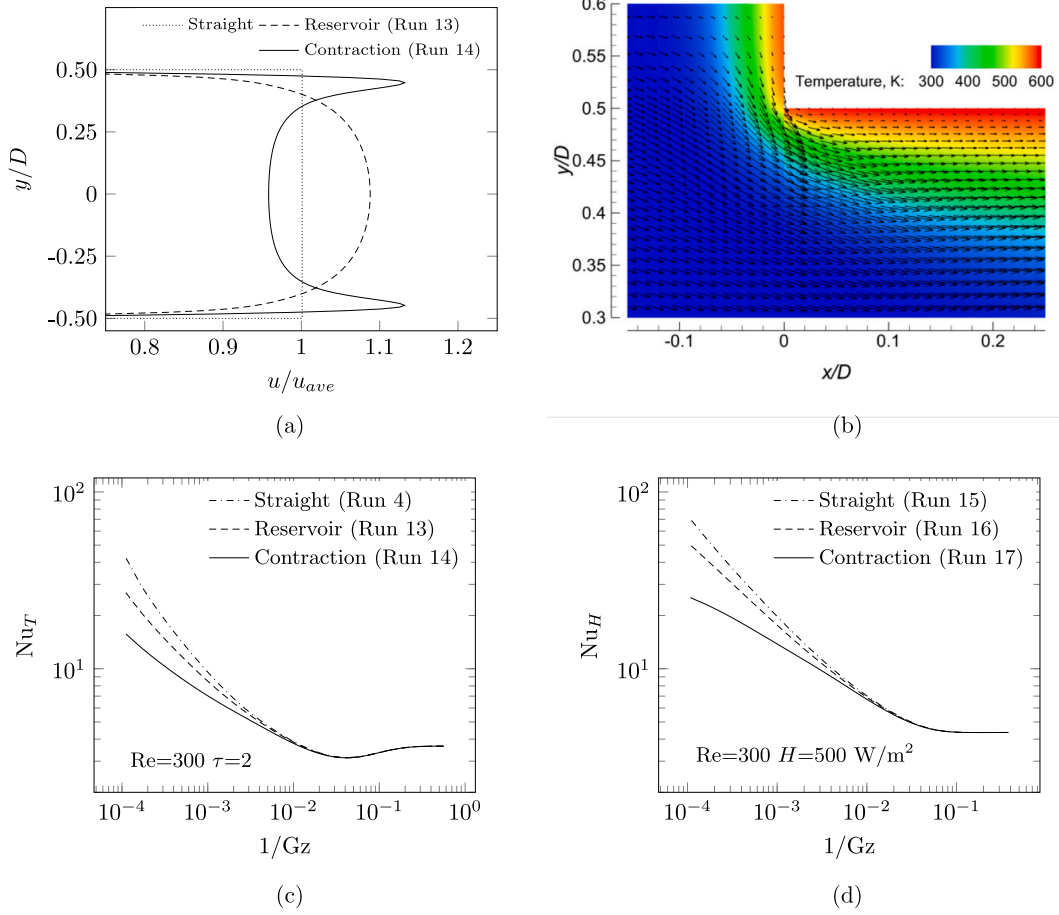


Fig. 6. (a) Compares the inlet velocity profile of a channel with the contraction and a channel with the reservoir. (b) is a contour plot of the temperature in a channel with the contraction at the inlet. (c) Nu_T and (d) Nu_H for the three channel shapes analysed. The run numbers refer to those in Table 2.

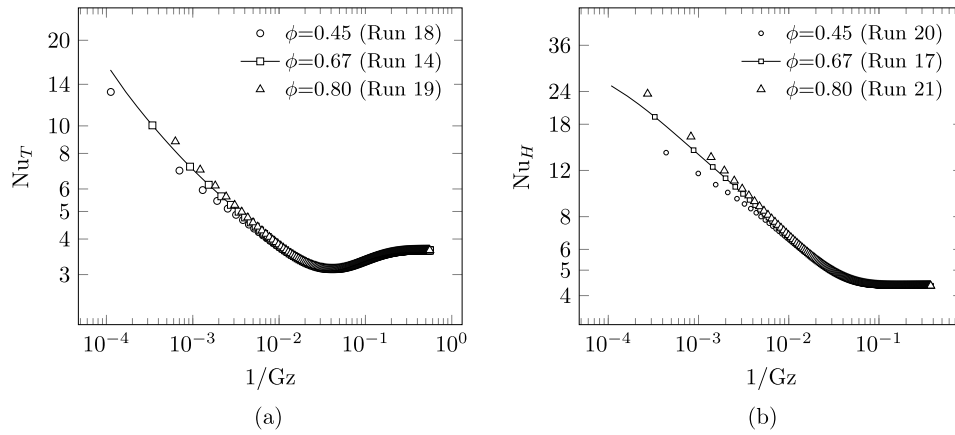


Fig. 7. Influence of the substrate void fraction on Nu. For all curves Re is 300 and T_i is 300 K. For Nu_T the wall temperature is 600 K and for Nu_H H is 500 kW/m². The run number refers to those in Table 2.

Unfortunately, a general solution could not be found and remains beyond the scope of this paper. However, the model was completed with empirical functions for $Nu_{T_{min}}$ and Gz_*^{-1} in Eq. (21) and (22), which can be reasonably assumed to be a function of τ only for the conditions covered in this study (see Fig. 8). Numerical data and the fitting curves for both variables are shown in Fig. 11.

$$Nu_{T_{min}} = 0.144264\tau^2 - 0.933583\tau + 4.433455 \quad (21)$$

$$Gz_*^{-1} = 0.002944\tau^2 - 0.000639\tau + 0.049663 \quad (22)$$

Empirical models for both $Nu_{T_{min}}$ and Gz_*^{-1} were provided because of their physical meaning. Mathematical expressions for the rest of the parameters, that is B , C , n , a and b , are also desired. However, because of the high non-linearity of the model and the considerable number of parameters, a comprehensive statistical analysis would be required, and thus a potential model reduction is highly recommended. These aspects will be addressed in a further paper.

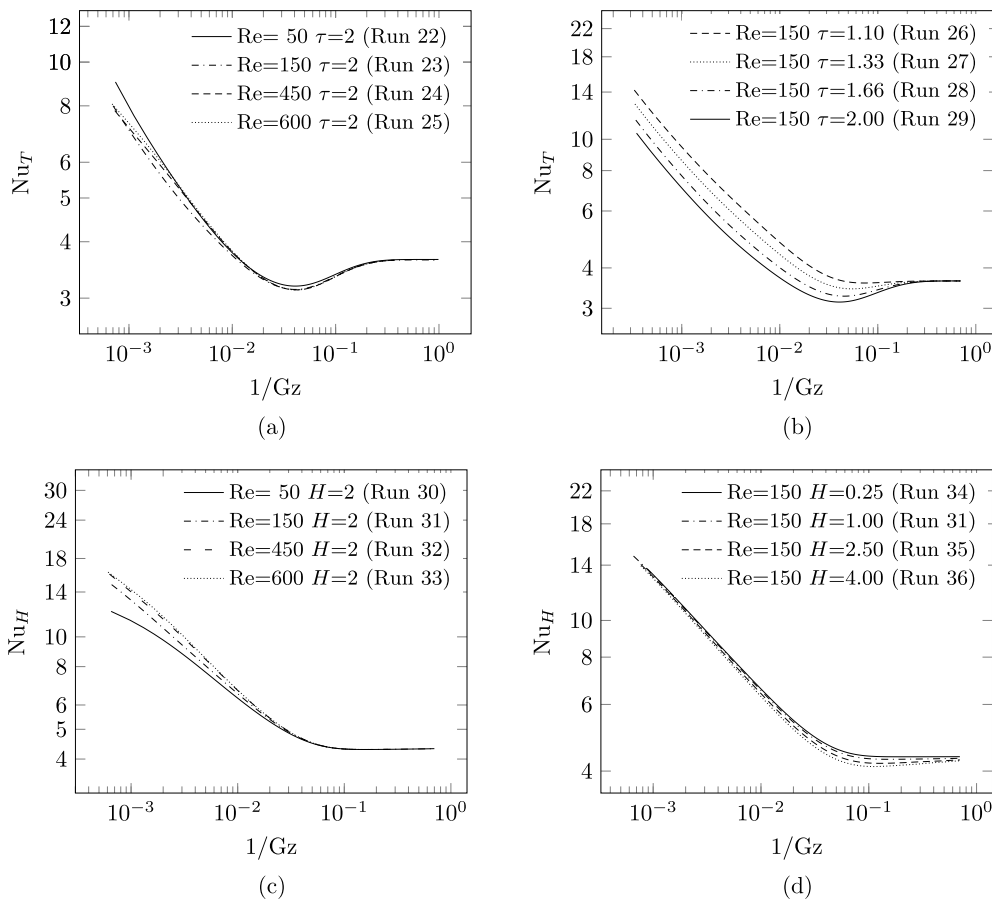


Fig. 8. Nu_T and Nu_H for several Re , τ and H (in kW/m^2). Effect of (a) Re in Nu_T at a constant τ , (b) τ in Nu_T at a constant Re , (c) Re in Nu_H at a constant H and (d) τ in Nu_H at a constant Re . The run number is referred to those in Table 2.

3.8. Summary of Nu expressions

Table 4 summarises the expressions for describing the Nu vs. Gz^{-1} developed in this paper. It can be appreciated that most complex case is that when the loading temperature is higher than 1.33. In such a case, most of the parameters of the functions are no longer constant, but a function of the loading temperature. For completeness and an easier implementation, empirical functions for B , C and n were also included in the table. The three mentioned expressions can be obtained straightforwardly by using the data in Table 3, and all of them show an R^2 of 0.99 or higher. Although, further research should be conducted in order to obtain more fundamental expressions.

4. Conclusions

Convective heat and mass transfer in monolith channels was successfully investigated using computational fluid dynamics.

Three geometrical representations of a monolith channel were studied: A straight channel, a channel with a reservoir prior to it and a channel with a contraction of the flow area at its inlet. The three of them led to a significantly different inlet velocity profile, where the channel with the contraction was the most realistic one. The three channel representations also led to a different thermal entrance length values in Nu vs. $1/Gz$ curve, where the channel with the contraction showed a lower Nu compared to the other two geometries. For the channel with a contraction, several substrate void fractions were analysed. According to the results, the inlet Nu increases together with the void fraction; however, the differences in the Nu vs. $1/Gz$ curve are small in magnitude and occur very close to the inlet of the channel only.

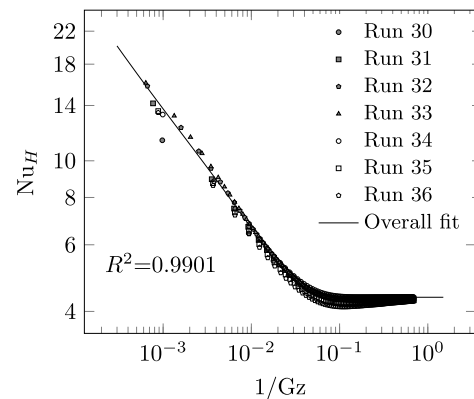


Fig. 9. Overall fitting curve to model Nu_H for several H and Re . The run number refers to those in Table 2.

The use of constant instead of temperature-dependent physical properties affected not only the resulting convective transport coefficient, but also the calculation of the non-dimensional axial coordinate. At a high heating rate, assuming constant fluid properties overestimates Nu over the entry length, and does not capture correctly the minimum that appears in the Nu vs. $1/Gz$. Regarding the non-dimensional axial distance, $1/Gz$ is largely overestimated at a high heating rate, in an error of the order of the 40%. At a τ as low as 1.1, the error in $1/Gz$ is of about 5%.

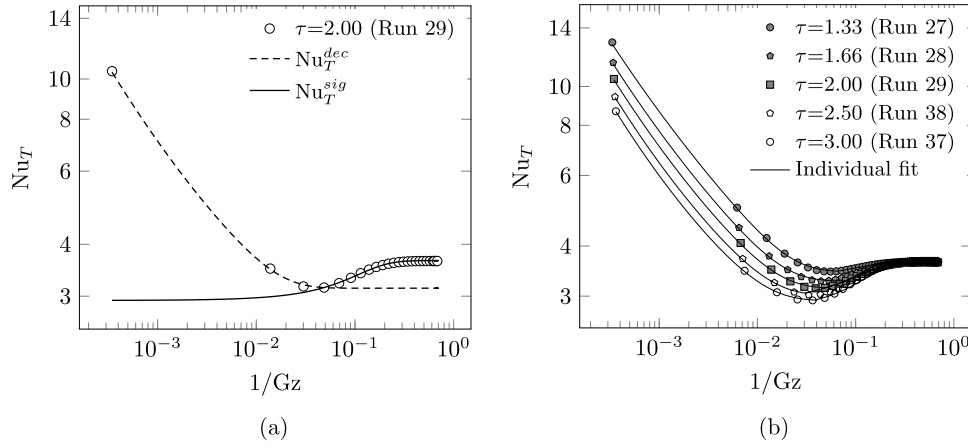


Fig. 10. (a) Nu_T^{dec} and Nu_T^{sig} curves fitted to a dataset. (b) Piece-wise function in Eq. (17) predicting Nu_T at several heating rates. The run number refers to those in Table 2.

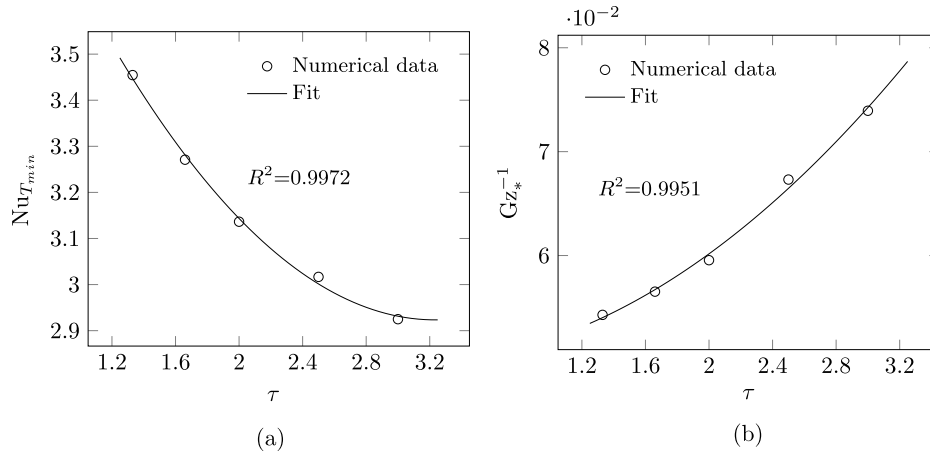


Fig. 11. Empirical polynomials for $Nu_{T,min}$ and Gz_*^{-1} as a function of τ .

Table 4

List of Nu expressions.

Constant wall heat flux	B	C	n
$Nu_H = Nu_{H_0} (1 + B(Gz)^n) \exp(-C/Gz)$	0.141	54.088	0.402
Constant wall temperature $\tau < 1.33$	B	C	n
$Nu_T = Nu_{H_0} (1 + B(Gz)^n) \exp(-C/Gz)$	0.0729	69.303	0.459
Constant wall temperature $\tau \geq 1.33$			
$Nu_T = \max(Nu_T^{dec}, Nu_T^{sig})$			
$Nu_T^{sig} = \left(\frac{Nu_{T_{\infty}}}{Nu_{T,min}} \right) \left[\left(\frac{1}{1+e^{-\zeta}} \right) + Nu_{T,min} - 1 \right]$			
$-\zeta = -0.502(Gz^{-1} - Gz_*^{-1}) + 15.575$			
$Nu_{T,min} = 0.144\tau^2 - 0.934\tau + 4.433$			
$Gz_*^{-1} = 0.00294\tau^2 - 0.000639\tau + 0.0497$			
$B = 0.0093\tau^2 - 0.0580\tau + 0.116$			
$C = 21.063\tau + 40.012$			
$n = -0.0132\tau^2 + 0.0948\tau + 0.3836$			

Over a certain range of τ a minimum noticeably below the asymptotic value appears in the Nu vs. $1/Gz$ curve. That is more pronounced at a higher heating rate, being about 16% lower than the asymptotic value for $\tau = 2$. The minimum is especially significant at a constant wall temperature and at a high heating rate; however, neither the placement

and the value of the minimum seems to be affected significantly by changes in Re . At a constant wall heat flux, the value of the minimum decreases just slightly with an increase of H , being always comparable to the asymptotic value and reasonably negligible under the conditions of this study.

Classical models are not able to describe Nu vs. $1/Gz$ curves with a minimum. Hence, a new model with a methodology to calibrate it was proposed. The new model mixes classical models with a sigmoid function used in machine learning to describe accurately the Nu vs. $1/Gz$ curve in all its extension, the decreasing zone, the concave part that contains the minimum and the asymptotic part showing an excellent agreement in every case. Empirical models for the value and the placement of the minimum Nu were also presented because of their physical meaning and relevance. Such models are a function of τ only because it resulted the most influential parameter for them, and apply for any of the Re covered in this study.

The same methodology presented in this paper can be used to investigate channels with other cross-section shapes commonly found in industry. Models accounting for other incremental effects, such as the one of the upstream turbulence, and flow entering the substrate in an angle can be investigated using the same methodology. A disadvantage of current models for Nu curves is the number of empirical parameters necessary to define to implement the model. A statistical analysis of parameter sensitivity and a potential model reduction are highly recommended to investigate further.

Nomenclature

Symbols

μ	Viscosity, Pa s
ϕ	Substrate void fraction
ρ	Density, kg/m ³
τ	Heating rate
B, C, n, a, b	Empirical parameters
C_p	Heat capacity, J/kg K
D	Channel diameter, m
D_1	Cell hydraulic diameter, m
D_a	Mass diffusivity, m ² /s
H	Wall heat flux, kW/m ²
h	Convective heat transfer coefficient, W/K m ²
k	Thermal conductivity, W/K m
L	Channel length, m
L_1	Upstream zone length, m
M_W	Molecular weight, g/mol
p	Pressure, Pa
R_g	Gas constant, J/mol K
T	Temperature, K
u_i	Velocity vector, m/s
x_i	Cartesian axis
Gz	Graetz Number = $RePrD/x$
Le	Lewis Number = Sc/Pr
Nu	Nusselt Number = hD/k
Pr	Prandtl Number = $C_p\mu/k$
Re	Channel Reynolds Number = $\rho uD/\mu$
Sc	Schmidt Number = $\mu/\rho D_a$
Sh	Sherwood Number = $h/D_a D$

Subscripts

∞	Asymptotic value
b	Bulk
H	Constant wall heat flux
i	Inlet (except for u_i and x_i)
m	For mass transfer
T	Constant wall temperature
w	Wall

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

I. Cornejo acknowledges the receipt of a Becas-Chile (CONICYT) scholarship. Other funding was provided by the Natural Sciences and Engineering Research Council of Canada. This research was enabled in part by support provided by Calcul Quebec, Canada (www.calculquebec.ca) and Compute Canada (www.computeCanada.ca).

References

- [1] Kazuhiro Yamamoto, Shingo Satake, Hiroshi Yamashita, Microstructure and particle-laden flow in diesel particulate filter, *Int. J. Therm. Sci.* 48 (2) (2009) 303–307.
- [2] Audrey Devatine, Hélène Chamaat, Simon Guillaume, Bismath Tati Tchibouanga, Freddy Durán Martínez, Carine Julcour, Anne-Marie Billet, Hydrodynamic study of a monolith-type reactor for intensification of gas-liquid applications, *Chem. Eng. Process.: Process Intensif.* 122 (2017) 277–287.
- [3] Aldo Javier Guadarrama-Mendoza, Heidi Isabel Villafán-Vidales, Patricio J Valadés-Pelayo, Camilo A Arancibia-Bulnes, David Riveros-Rosas, Hernando Romero-Paredes, Radiative analysis in a multichanneled monolith solar reactor coated with ZnFe₂O₄ thin film, *Int. J. Therm. Sci.* 132 (2018) 275–284.
- [4] Piyanut Inbamrung, Thana Sornchamni, Chaiwat Prapainainar, Sabaithip Tungkaman, Phavanee Narataruksa, Goran N Jovanovic, Modeling of a square channel monolith reactor for methane steam reforming, *Energy* (2018).
- [5] Nur Izwanne Mahyon, Tao Li, Ricardo Martinez-Botas, Zhentao Wu, Kang Li, A new hollow fibre catalytic converter design for sustainable automotive emissions control, *Catal. Commun.* 120 (2019) 86–90.
- [6] Selvakumar Kumares, Man Young Kim, Numerical investigation of catalytic combustion in a honeycomb monolith with lean methane and air premixtures over the platinum catalyst, *Int. J. Therm. Sci.* 138 (2019) 304–313.
- [7] L. Graetz, Über die Wärmeleitungsfähigkeit Von Flussigkeiten—Part 1, *Ann. Phys. Chem.* 18 (1883) 79–94.
- [8] W. Kays, M. London, *Compact Heat Exchangers*, McGraw-Hill, New York, NY, USA, 1964.
- [9] U. Grigull, H. Tratz, Thermischer einlauf in ausgebildeter laminarer rohrströmung, *Int. J. Heat Mass Transfer* 8 (5) (1965) 669–678.
- [10] Enrico Tronconi, Pio Forzatti, Adequacy of lumped parameter models for SCR reactors with monolith structure, *AIChE J.* 38 (2) (1992) 201–210.
- [11] Robert E. Hayes, Stan T. Kolaczkowski, *Introduction to Catalytic Combustion*, CRC Press, Boca Raton, FL, USA, 1998.
- [12] Robert E. Hayes, Andrés Donoso-Bravo, Joseph P. Mmbaga, Entry length effects for momentum, heat and mass transfer in circular ducts with laminar flow, *Can. J. Chem. Eng.* 93 (5) (2015) 863–869.
- [13] C.F. Bonilla, *Nuclear Engineering Handbook*, McGraw-Hill, New York, NY, USA, 1958.
- [14] Renate Schwiedernoch, Steffen Tischer, Chrys Correa, Olaf Deutschmann, Experimental and numerical study on the transient behavior of partial oxidation of methane in a catalytic monolith, *Chem. Eng. Sci.* 58 (3–6) (2003) 633–642.
- [15] Heinz Herwig, Shripad P. Mahulikar, Variable property effects in single-phase incompressible flows through microchannels, *Int. J. Therm. Sci.* 45 (10) (2006) 977–981.
- [16] R.W. Shumway, D.M. McEligot, Heated laminar gas flow in annuli with temperature-dependent transport properties, *Nucl. Sci. Eng.* 46 (3) (1971) 394–407.
- [17] JM Nóbrega, Fernando Tavares de Pinho, Paulo J Oliveira, OS Carneiro, Accounting for temperature-dependent properties in viscoelastic duct flows, *Int. J. Heat Mass Transfer* 47 (6–7) (2004) 1141–1158.
- [18] R.E. Hayes, B. Liu, R. Moxom, M. Votsmeier, The effect of washcoat geometry on mass transfer in monolith reactors, *Chem. Eng. Sci.* 59 (15) (2004) 3169–3181.
- [19] Oswald Jason Lobo, Dhiman Chatterjee, Development of flow in a square mini-channel: Effect of flow oscillation, *Phys. Fluids* 30 (4) (2018) 042003.
- [20] Ivan Cornejo, Petr Nikrityuk, Robert E. Hayes, Pressure correction for automotive catalytic converters: A multi-zone permeability approach, *Chem. Eng. Res. Des.* 147 (2019) 232–243.
- [21] I. Cornejo, P. Nikrityuk, R.E. Hayes, Turbulence decay inside the channels of an automotive catalytic converter monolith, *Emiss. Control Sci. Technol.* 3 (4) (2017) 302–309.
- [22] Ivan Cornejo, Petr Nikrityuk, Robert E. Hayes, Multiscale RANS-based modeling of the turbulence decay inside of an automotive catalytic converter, *Chem. Eng. Sci.* 175 (2018) 377–386.
- [23] Ivan Cornejo, Robert E. Hayes, Petr Nikrityuk, A new approach for the modeling of turbulent flows in automotive catalytic converters, *Chem. Eng. Res. Des.* 140 (2018) 308–319.
- [24] Ivan Cornejo, Gonzalo Cornejo, Petr Nikrityuk, Robert E Hayes, Entry length convective heat transfer in a monolith: The effect of upstream turbulence, *Int. J. Therm. Sci.* 138 (2019) 235–246.
- [25] ANSYS Fluent Theory Guide v18.2, ANSYS Inc., Canonsburg, PA, USA, 2017.
- [26] Bruce E Poling, John M Prausnitz, John P O'connell, et al., *The Properties of Gases and Liquids*, Vol. 5, McGraw-hill New York, 2001.
- [27] ANSYS Fluent Software Package v18.2, ANSYS Inc., Canonsburg, PA, USA, 2017.
- [28] W.M. Kays, M.E. Crawford, Bernhard Weigand, *Convective Heat and Mass Transfer*, McGraw-Hill Higher Education, New York, NY, USA, 2004.
- [29] M.S. Bhatti, Fully developed temperature distribution in a circular tube with uniform wall temperature, 1985, unpublished.
- [30] Theodore L Bergman, Frank P Incropera, David P DeWitt, Adrienne S Lavine, *Fundamentals of Heat and Mass Transfer*, John Wiley & Sons, New Jersey, NY, USA, 2011.
- [31] H. More, J. Mmbaga, R.E. Hayes, M. Votsmeier, M.D. Checkel, Heat and mass transfer limitations in pre-turbocharger catalysts, *Top. Catalysis* 42 (1–4) (2007) 429–432.
- [32] Yi Liu, Michael P. Harold, Dan Luss, Coupled NO_x storage and reduction and selective catalytic reduction using dual-layer monolithic catalysts, *Appl. Catal. B* 121 (2012) 239–251.
- [33] Ivan Cornejo, Petr Nikrityuk, Robert E. Hayes, The influence of channel geometry on the pressure drop in automotive catalytic converters: Model development and validation, *Chem. Eng. Sci.* (2020) <http://dx.doi.org/10.1016/j.ces.2019.115317>.